

Real-Time ASL Alphabet Recognition

Project Report | Elevate Labs AI/ML Internship

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Abstract

This project presents a real-time American Sign Language (ASL) alphabet recognition system that converts hand gestures (A-Z) into text from a live webcam feed. The pipeline extracts 21 hand landmarks per frame using the MediaPipe Tasks API, normalizes them for position and scale invariance, and classifies them using a 1D-CNN (TensorFlow/Keras) and an MLP (scikit-learn). A LetterStabilizer debounce logic ensures accurate letter commitment, while a SentenceBuilder combines letters into words with auto-spacing and optional text-to-speech output. The system achieves 99.8% test accuracy on 10,508 samples across 26 classes.

Introduction

American Sign Language (ASL) is the primary visual language used by the Deaf and hard-of-hearing community. Real-time gesture-to-text translation can bridge communication gaps and improve accessibility. This project aims to develop a lightweight, accurate system that recognizes static ASL alphabet signs (A-Z) from live webcam video and converts them into readable text. Unlike conventional image-based approaches, this system uses pre-extracted MediaPipe landmarks (63 numeric features per frame), significantly reducing computational overhead while maintaining near-perfect accuracy.

Tools Used

- Python 3.10+ - Core development and orchestration
- OpenCV - Webcam capture, frame rendering, and UI overlay
- MediaPipe Tasks API - 21-point (x, y, z) hand landmark detection
- TensorFlow / Keras - 1D-CNN model training and TFLite edge export
- scikit-learn - MLPClassifier, StandardScaler, and LabelEncoder
- NumPy - Landmark array operations, normalization, and augmentation
- joblib - Model serialization (MLP + scaler)
- pyttsx3 - Optional text-to-speech output (~150 wpm)
- Matplotlib / Seaborn - Confusion matrix and skeleton visualization

Steps Involved in Building the Project

1. Dataset Loading & Exploration

Loaded 10,508 samples from the ASL Alphabet Hand Landmarks dataset (26 balanced classes, A-Z). Each sample stores 21 MediaPipe landmarks as .npy files with (x, y, z) coordinates.

2. Preprocessing & Normalization

Applied `normalize_landmarks()` to make data position and scale-invariant: translated so the wrist (landmark 0) sits at the origin, then rescaled by the maximum joint distance. This corrected a dataset quality issue where only the Z coordinate was truly invariant. Features were flattened to 63-D vectors.

3. Model Development

Two architectures were trained on identical normalized features: (a) a 1D-CNN treating landmarks as a (21, 3) grid with Conv1D layers (64->128->128), batch normalization, and dropout; and (b) an MLP quickstart (256->128->64) with ReLU activations, early stopping, and StandardScaler. The MLP achieved 99.8% test accuracy with only 3 misclassifications (D->O, R->U).

4. Real-Time Inference Pipeline

Built `webcam_recognition.py` using MediaPipe's HandLandmarker in VIDEO mode. The script captures frames, extracts landmarks, applies the same normalization used during training, and predicts letters via the trained model. A dual-backend predictor supports both the Keras CNN and the scikit-learn MLP.

5. Debounce Logic & Sentence Building

Implemented LetterStabilizer requiring 15 consecutive frames above 75% confidence before committing a letter, filtering

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jitter and transition noise. SentenceBuilder adds auto-spacing after 45 frames of no hand, supports double-letter fingerspelling via release frames, and provides backspace, clear, and text-to-speech controls.

6. Evaluation & Visualization

Generated a 26x26 confusion matrix, 3D/2D skeleton plots for Letter A, and an offline demo video replaying real held-out samples through the full inference pipeline to demonstrate genuine predictions without requiring a webcam.

Conclusion

The project successfully demonstrates that landmark-based ASL recognition achieves high accuracy (99.8%) with minimal computational cost, making it suitable for real-time edge deployment. The wrist-centered normalization step proved critical for generalization across varying hand positions and sizes. The debounce logic transforms noisy per-frame predictions into a clean, usable text stream. Limitations include single-signer dataset bias, near-duplicate train/test frames inflating accuracy, and static-only sign recognition (J and Z lack motion cues). Future work should expand to dynamic gestures, multi-signer datasets, and full-word ASL sign recognition to create a more comprehensive communication aid.